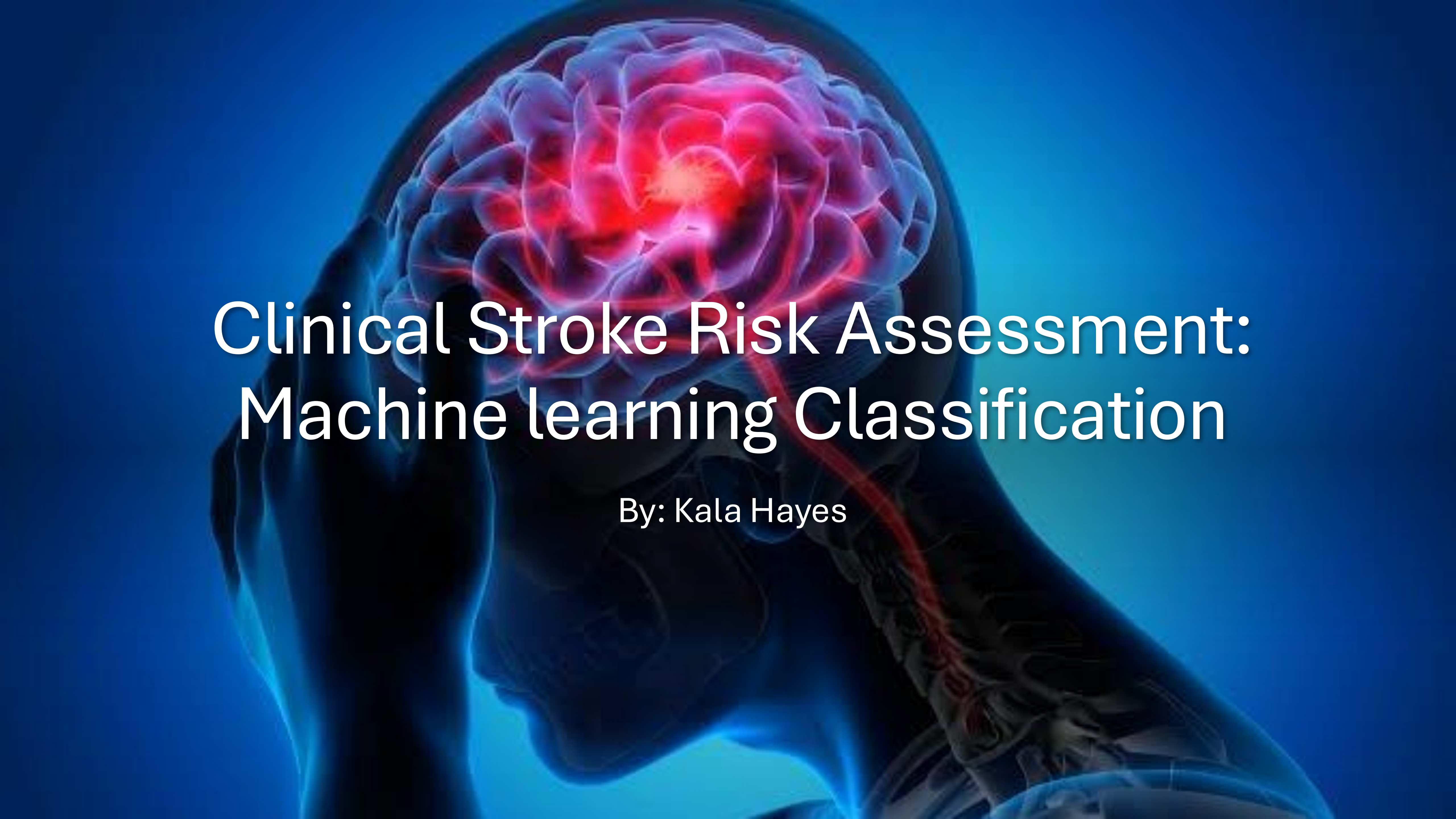

Clinical Stroke Risk Assessment: Machine learning Classification

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Clinical Stroke Prediction Using Machine Learning

Project Goal

Develop machine learning models to predict stroke outcomes using patient clinical and demographic data.

Dataset Challenge

- Highly imbalanced dataset:
  - 95.15% non-stroke cases
  - 4.85% stroke cases
- Accuracy alone was not enough to evaluate model performance.

Models Evaluated

- Logistic Regression
- Decision Tree
- Random Forest
- Gradient boosting
- KNN
- SVC

Evaluation

Accuracy, Precision, Recall, F1-score, ROC-AUC

Data Preparation & Workflow

Data Processing

- Split Data First: Established70%|15%|15% partitioning upfront to prevent leakage
- Missing BMI values handled with data- confined median imputation
- Numerical features standardized around zero-mean variance boundaries
- Categorical features encoded using one-hot expansion

Class Imbalance Handling

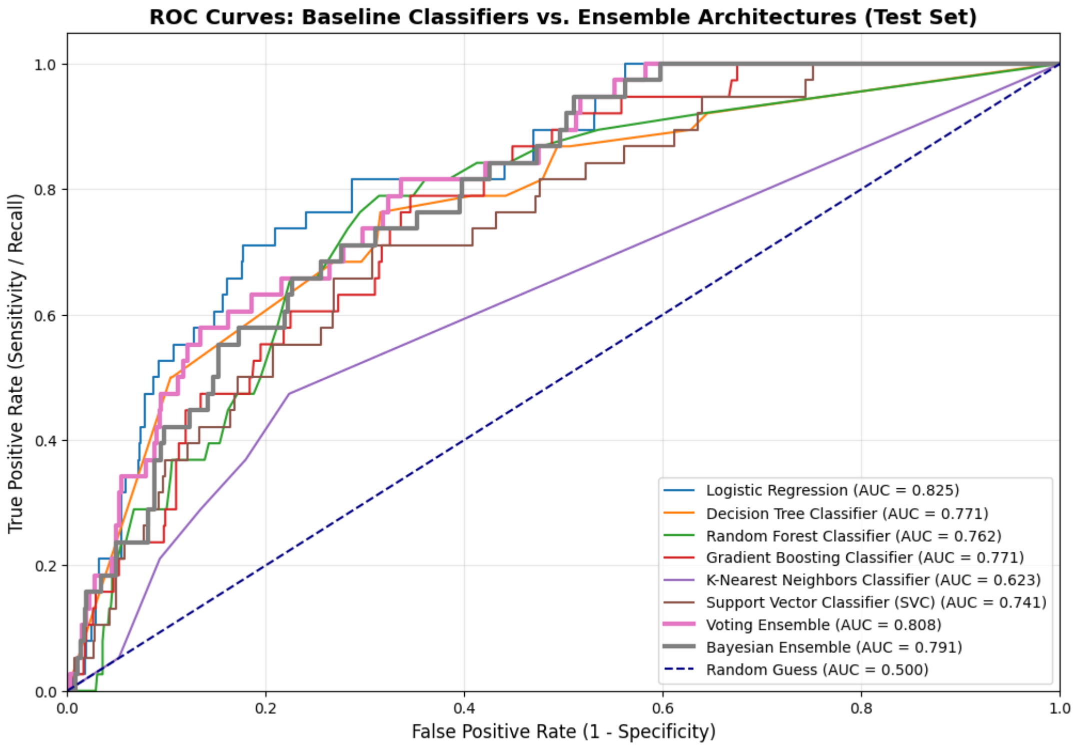
  - SMOTE was applied only to the training subsets only to balance the target distribution
  - Validation and test data remained completely raw and untouched by sampling

Workflows Sequence

Split Data-->Preprocess --> Train Models--> Compare Results --> Build & Evaluate Ensembles

Model Performance Results

- **Key Finding**
- Random Forest recorded a high raw test accuracy of 91.66%, but yielded an abysmal minority class accuracy of 13.16%. showing why accuracy alone was not enough. This exposes the majority class bias fallacy and confirms why accuracy alone is a deceptive evaluation metric.



Model	ROC-AUC	Accuracy	Recall
Logistic Regression	0.825	71.71%	81.58%
Random Forest	0.762	91.66%	13.16%
Voting Ensemble	0.808	74.71%	65.79%
Bayesian Ensemble	0.791	80.83%	57.89%

Ensemble Comparison & Conclusion

Model	Accuracy	F1	ROC-AUC
Voting Ensemble	74.71%	0,2049	0.808
Bayesian Ensemble	80.83 %	0.2304	0.791

What I learned

- Bayesian Ensemble --> Achieved the peak ensemble accuracy and F1-score (0.2304)
- Voting Ensemble -->Generated a smoother probability aggregation for a better ensemble ROC-AUC
- Multi-model aggregation stabilized predictive variance across borderline clinical thresholds, though standalone Logistic Regression maintained the highest overall class separation matrix boundaries.

Final Takeaway

For clinical stroke prediction systems, model selection must prioritize the operational metric that best supports the overarching clinical goal of minimizing patient risk. Standalone Logistic Regression's high recall (81.58%) makes it exceptionally valuable for diagnosing true positive events, successfully preventing critical false negatives.